

Day 8: SQL Notes Basic to Advanced

Today's Topic: SQL Aggregate Functions

- COUNT()
- SUM()
- AVG()
- MIN()
- MAX()

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SQL Aggregate Functions

- SQL aggregate functions are used to perform calculations on a set of values and return a single value.
- These functions are commonly used with the GROUP BY clause to group rows that have the same values in specified columns into summary rows.
- Perform calculations on multiple rows of data to return a single result.
- Often used with GROUP BY to summarize data by groups.
- Useful for generating summary statistics from large datasets.

Consider the two tables below:

EmployeeID	Name	Department	Salary	JoinDate
1	John Doe	IT	60000	2020-01-15
2	Jane Smith	HR	50000	2019-03-22
3	Michael Lee	IT	75000	2018-07-11
4	Emily Davis	Marketing	45000	2021-06-19
5	James Brown	IT	65000	2022-02-10
6	Susan Clark	HR	52000	2020-09-05
7	Robert King	Marketing	48000	2019-12-01

8.1 Count()

Returns the number of rows that match a specified condition, including NULL values when using **COUNT(*)**.

- Calculates the total number of rows in a result set.
- Ignores NULL values unless specifically counted.
- Commonly used with **GROUP BY** to count rows within groups.
- Useful for determining the size of a dataset or specific criteria.

Example

```
SELECT SUM(Salary)
FROM Employees;
```

Output

```
COUNT(*)
7
```

Explanation: The `COUNT(*)` function counts all rows in the `Employees` table, which totals 7 employees.

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8.2 Sum()

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Description: Adds up all the values in a numeric column, excluding NULL values.

- Calculates the total of numeric values in a specified column.
- Often used with `GROUP BY` to total values by category.
- Ignores NULL values in the calculation.
- Helps analyze data, like finding total sales or expenses.

Use Case: Useful for financial or sales data, like calculating total sales.

Example:

Calculate the total salary paid to all employees.

```
SELECT SUM(Salary)
FROM Employees;
```

Output

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```
SUM(Salary)
395000
```

Explanation: The `SUM(Salary)` function adds up all the salary values from the `Employees` table, resulting in a total of 395,000.

8.3 Avg()

The `AVG()` function calculates the average value of a numeric column.

- Calculates the average value of a numeric column.
- Ignores NULL values in the calculation.
- Commonly used with the `GROUP BY` clause to get averages for groups.
- Useful for summarizing data, like finding the average salary or rating.

Example:

Calculate the average salary of all employees.

```
SELECT AVG(Salary)
FROM Employees;
```

Output

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```
AVG(Salary)
56428.57
```

Explanation: The `AVG(Salary)` function calculates the average of the salary values, which is approximately 56,428.57. This is useful for understanding the overall compensation level in the company.

8.4 Min()

The `MIN()` function returns the minimum value in a set of values.

- Returns the smallest value in a specified column.
- Can be used with numeric, date, and string data types.
- Ignores NULL values in the column.
- Often used in combination with the `GROUP BY` clause to find minimum values for each group.

Example:

Find the minimum salary among all employees.

```
SELECT MIN(Salary)
FROM Employees;
```

Output

```
MIN(Salary)
45000
```

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Explanation: The `MIN(Salary)` function identifies the lowest salary in the `Employees` table, which is 45,000. This helps to identify the base level of pay within the organization.

8.5 Max()

The `MAX()` function returns the maximum value in a set of values.

- Returns the highest value from a specified column.
- Operates on numerical, date, and string data types.
- Ignores NULL values when determining the maximum.
- Often used with `GROUP BY` to find maximum values within groups.

Example:

Find the maximum salary among all employees

```
SELECT MAX(Salary)
FROM Employees;
```

Output

```
MAX(Salary)
75000
```

Explanation: The `MAX(Salary)` function finds the highest salary in the `Employees` table, which is 75,000. This is useful for identifying the top earners or the upper limit of the salary structure.

Using Aggregate Functions with GROUP BY

The `GROUP BY` clause is used in conjunction with aggregate functions to group the result set by one or more columns.

Example:

Calculate the total salary paid in each department.

```
SELECT Department, SUM(Salary)
FROM Employees
GROUP BY Department;
```

Output

Department	SUM(Salary)
HR	102000
IT	200000
Marketing	93000

Explanation

- This query groups the data by the Department column and calculates the total salary for each department.
- The **GROUP BY** clause ensures that the **SUM(Salary)** function is applied separately to each department, resulting in the total salaries of 102,000 for HR, 200,000 for IT, and 93,000 for Marketing.

Using Aggregate Functions with HAVING

The HAVING clause is used to filter groups based on the results of aggregate functions.

Example

Find departments where the total salary is greater than 100,000.

```
SELECT Department, SUM(Salary)
FROM Employees
GROUP BY Department
HAVING SUM(Salary) > 100000;
```

Output

Department	SUM(Salary)
IT	200000
HR	102000

- The HAVING clause filters the results of the GROUP BY operation to only include groups (departments) where the total salary is greater than 100,000.
- In this case, only the IT and HR departments meet this criterion, with total salaries of 200,000 and 102,000 respectively.

Using Aggregate Functions with HAVING

The HAVING clause is used to filter groups based on the results of aggregate functions.

Example

Find departments where the total salary is greater than 100,000.

```
SELECT Department, SUM(Salary)
FROM Employees
GROUP BY Department
HAVING SUM(Salary) > 100000;
```

Output

Department	SUM(Salary)
IT	200000
HR	102000

- The HAVING clause filters the results of the GROUP BY operation to only include groups (departments) where the total salary is greater than 100,000.
- In this case, only the IT and HR departments meet this criterion, with total salaries of 200,000 and 102,000 respectively.

Day 9:SQL Views

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